Let us summarise the working rule to solve an oblique triangle in the following table

Oblique triangle (all angles are acute or one angle is obtuse	Given Information				
	SAA	SSA*	SAS	SSS	AAA
Details and Application for solutions	Law of sines	#(ambiguity arises)	The given angle must be included angle; Either law of Cosines or tangents	Law of Cosines; First find the largest angle	Infinite number of triangles

\# Angle is not included; We may have more than one triangle;
Application of law of sines yields three cases: No solution or one triangle or two triangles.
Suppose a, b, A are known. Let $h = b\sin A$
If $a < h$, there is no triangle, If $a = h$, then it is right triangle.
If $a > h$ and $a < b$, we have two triangles.
If $a \geq b$, we have only one triangle.

* SSA means Side, Side and Angle

Example 3.64 In a $\triangle ABC$, $a = 3, b = 5$ and $c = 7$.
Find the values of $\cos A$, $\cos B$ and $\cos C$.

Solution:
By Cosine formula,

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{5^2 + 7^2 - 3^2}{2(5)(7)} = \frac{13}{14}$$

Similarly, $\cos B = \frac{11}{14}$, $\cos C = -\frac{1}{2}$.

Example 3.65 In $\triangle ABC$, $A = 30°, B = 60°$ and $c = 10$, Find a and b.

Solution:
Given that $A = 30°, B = 60°, C = 180° - (A + B) = 90°$
Using sine formula,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\frac{a}{\sin 30°} = \frac{b}{\sin 60°} = \frac{10}{\sin 90°}$$

$$a = \frac{10\sin 30°}{\sin 90°} = \frac{10\left(\frac{1}{2}\right)}{1} = 5$$

$$b = \frac{10\sin 60°}{\sin 90°} = \frac{10\left(\frac{\sqrt{3}}{2}\right)}{1} = 5\sqrt{3}.$$

Example 3.66 In a $\triangle ABC$, if $a = 2\sqrt{2}, b = 2\sqrt{3}$ and $C = 75°$, find the other side and the angles.

Solution:
Given that $a = 2\sqrt{2}, b = 2\sqrt{3}$ and $C = 75°$.